GOWTHAM PUGALENTHI

AI/ML Engineer | Full-Stack Developer | Dharmapuri, Tamil Nadu, India
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AI/ML Engineer (B.Tech, final year) specializing in deep learning, NLP, anomaly detection, and generative AI. Built and deployed end-to-end ML pipelines and full-stack applications with measurable production outcomes. Proficient in PyTorch, TensorFlow, Keras, RASA, and the MERN stack. Recognized at GenAI Competition 2024 (1st place) and StartupTN.

TECHNICAL SKILLS

AI/ML: Deep Learning (CNN, RNN, GAN, Transformers) | NLP (BERT, TF-IDF, SVM, RASA) | Anomaly Detection | Feature Engineering | Transfer Learning | XGBoost | Scikit-learn
Languages & Frameworks: Python, JavaScript, PyTorch, TensorFlow, Keras, React.js, Node.js, Next.js, Express.js, Laravel
Tools & Platforms: Git, MongoDB, MySQL, Hugging Face, Postman, Google Colab, Kaggle, JWT, Redux, Stripe

PROFESSIONAL EXPERIENCE

Younderbots | *Machine Learning Specialist* *Mar 2026 – July 2026*

- Designed and deployed ML models for business use cases, integrating deep learning pipelines that automated data processing and reduced manual workflows by **60%**.
- Built NLP-based text classification and extraction modules using BERT fine-tuning; improved model inference latency for production deployment.

Aakam360 | *Software Developer* *Jan 2025 – Mar 2026*

- Developed 5+ full-stack applications using Next.js and MERN stack; reduced application load time by **35%** through code-splitting, lazy loading, and query optimization.
- Implemented JWT-based authentication and role-based access control, strengthening application security across all deployed services.

FintechGie | *Web Development Intern* *Jan 2024 – Dec 2024*

- Engineered a knowledge assessment platform; reduced API latency by **40%** via query optimization and caching, contributing to a **30%** lift in user engagement.
- Built **Young Chanakya** CMS using Laravel integrated with a payment gateway, delivered and launched in production.

PROJECTS

Credit Card Fraud Detection System GitHub

- *What:* Real-time anomaly detection pipeline for fraudulent transaction identification on imbalanced financial data.
- *How:* Applied Isolation Forest and XGBoost with SMOTE oversampling; engineered 15+ domain-specific transactional features for signal extraction.
- *Impact:* Achieved 99.2% detection accuracy with <2% false positives, reducing simulated fraud loss by **95%** on test dataset.

Conversational AI Chatbot (RASA Framework) GitHub

- *What:* Context-aware NLP chatbot supporting multi-turn dialogue and intent recognition across 50+ intents.
- *How:* Built custom RASA pipeline with entity extraction, slot filling, and response ranking; fine-tuned intent classifier on domain-specific corpus.
- *Impact:* Handled **500+ conversations/month** autonomously; achieved **94% intent accuracy**, eliminating need for human-in-the-loop triage.

AI-Powered Email Classifier GitHub

- *What:* NLP system for automatic routing of incoming customer support emails into structured ticket categories.
- *How:* Curated a labeled dataset of 2,000+ emails; trained TF-IDF + SVM classifier with domain-specific vocabulary and feature selection.
- *Impact:* Reduced manual triage time by **70%** and improved SLA compliance to **98%** through automated ticket routing.

Generative Adversarial Network – Fashion Image Synthesis (TensorFlow/Keras) GitHub

- *What:* Built a GAN from scratch to generate synthetic 28×28 grayscale clothing images using the Fashion-MNIST dataset.
- *How:* Designed a Generator using Dense, UpSampling2D, and Conv2D layers with LeakyReLU activations; implemented a custom training loop to jointly optimize Generator and Discriminator.
- *Training:* Monitored Generator and Discriminator losses across epochs to evaluate convergence and identify training instability and potential mode-collapse behavior.

Wildfire Prediction System GitHub

- *What:* Regression model to forecast forest fire spread intensity from meteorological and temporal inputs.
- *How:* Engineered 12 features from weather/time data; trained XGBoost and Ridge ensemble with hyperparameter tuning via GridSearchCV.
- *Impact:* Attained R-Square of **0.91** on held-out test set; model outputs directly usable for fire risk scoring and early-warning thresholds.

E-Commerce Platform with AI Recommendations (MERN) GitHub

- *What:* Full-stack marketplace with integrated collaborative-filtering recommendation engine.
- *How:* Built React + Redux frontend, Node/Express backend with JWT auth and Stripe payment integration; implemented user-item collaborative filtering for personalization.
- *Impact:* Processed **r**ecommendation engine drove a **25% simulated uplift** in click-through rate on suggested products.

EDUCATION & CERTIFICATIONS

B.Tech – Artificial Intelligence & Data Science | Sri Shanmugha College of Engineering and Technology *Aug 2023 – Present*
CGPA: **8.1/10** | **Achievements:** 1st Place – GenAI Competition 2024 | 2nd Place – Debugging Competition 2024 | StartupTN Recognition
Certifications: Complete Data Science, ML, DL & NLP Bootcamp – Udemy (99 hrs, 2025) | Generative AI & Prompt Engineering (2023) | Python Problem Solving (2024)